

Machine-learning-based model for analysing and accurately predicting factors related to burnout in healthcare workers

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ABSTRACT

Objective The aim of this study is to analyse the factors affecting medical burnout in hospitals, identify the characteristics of staff experiencing high levels of burnout and devise a practical and sustainable prediction mechanism.

Methods A survey was conducted to access the current situation, followed by a regression analysis using data from the Maslach Burnout Inventory General Survey, demographic information related to healthcare personnel and employee job satisfaction metrics from the hospitals under study. Subsequently, four predictive models—logistic regression, K-nearest neighbour, decision tree and random forest (RF)—were employed to predict the degree of healthcare burnout.

Results The investigation revealed that 61.2% of the medical staff in the hospitals under study exhibited at least one symptom of burnout, with 9.8% experiencing high levels of burnout. Elevated rates of high burnout were observed in the 30–39 age group, among physicians and surgeons, and among those with 0–5 years of experience. In terms of predictive methods, the RF model demonstrated suitability for predicting burnout among medical staff, achieving a prediction accuracy of approximately 80%.

Conclusions A significant correlation was found between job satisfaction and burnout levels. Physicians and surgeons with less than a decade of professional experience are more prone to high levels of burnout. The RF model proved effective for predicting the burnout level among medical staff, consistently achieving an accuracy rate close to 80%. These findings can serve as valuable insights for hospital administrators in their effort to prevent and mitigate burnout among medical staff.

INTRODUCTION

Medical burnout is a growing issue among healthcare professionals worldwide. Recent reports show an alarming increase in burnout rates.^{1–3} For instance, the 2022 Physician Burnout Report published by Medscape, which surveyed more than 13 000 physicians across different specialties, revealed that nearly half of all doctors reported

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Medical staff burnout is a social problem that needs urgent attention and improvement in the global healthcare industry. Early prediction and prevention can improve medical burnout. Although there are many studies on the impact of medical burnout, there are few predictive models and preventive mechanisms that are practical and sustainable.

WHAT THIS STUDY ADDS

⇒ A significant correlation between job satisfaction and burnout levels. Physicians and surgeons with less than 10 years of experience were more likely to experience high levels of burnout. The random forest model was effective in predicting the level of burnout among medical professionals with an accuracy of nearly 80%.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These findings can serve as valuable insights for hospital administrators in their effort to prevent and mitigate burnout among medical staff.

experiencing burnout in 2021, an increase of 5% compared with 2020.⁴ A hospital-based cross-sectional survey of 25 000 medical staff in six provinces of China indicated that 60.8% of the medical staff exhibited at least one symptom of burnout.⁵

Burnout can have serious consequences. It is strongly associated with emotional exhaustion and depersonalisation, which can lead to alcohol abuse or dependence.⁶ A survey by Mayo Clinic in 2012 also found that doctor burnout increases the risk of traffic accidents.⁷ Furthermore, burnout can lead to increased levels of occupational pain and higher risk of suicide among doctors.^{8,9} Burnout not only severely affects the physical and mental health of medical staff but also impacts patient safety and healthcare organisations.^{10,11} A systematic

review found a significant association between medical staff burnout and patient safety.¹² The American College of Surgeons has indicated that major medical errors reported by surgeons are strongly correlated with their levels of burnout.¹³

Given the serious implications of medical burnout, it is a pressing issue in the global healthcare industry that requires immediate attention and improvement. Therefore, it is crucial to investigate and predict medical burnout. In this study, we will analyse the factors affecting burnout among medical staff in hospitals by combining the results of burnout surveys, employee satisfaction and demographic characteristics. In addition, we also identify the characteristics of burnout among high-level medical staff and design a practical and sustainable prediction mechanism using various machine-learning methods.

MATERIALS AND METHODS

Study design and data collection

This study conducted an anonymous cross-sectional survey of healthcare employees at Taizhou Hospital in Zhejiang Province. The researchers met the participants, explained the purpose of the study, the risks and benefits of participation and their right to refuse participation. They emphasised that the survey was voluntary, the results would remain anonymous and that there were no right or wrong answers. To calculate the appropriate sample size, a z value of 1.96 was used at the 95% confidence level. Based on the literature, the burnout rate of medical personnel in China is between 60% and 65%. Therefore, we considered a burnout rate of 60% with an acceptable error limit of 0.05.^{5 14} This calculation yielded a sample size of not less than 369.^{15 16} The data collection period spanned from 20 October to 20 November 2020. A total of 1125 valid samples were included, yielding a response rate of 66.6% (1125/1690). All procedures adhered to the guidelines of the Institutional Review Board of Taizhou Hospital of Zhejiang Province, affiliated with Wenzhou Medical University (approval number: K20230615), and aligned with the tenets of the Declaration of Helsinki. All participant information was collected anonymously.

Measurement

Maslach burnout inventory general survey (MBI-GS)

In this study, we used the Chinese version of the MBI-GS modified by Li, which consists of three parts: emotional exhaustion (EE, five items), depersonalisation (DP, four items) and personal accomplishment (PA, six items); the internal consistency coefficients of the three dimensions were 0.88, 0.83 and 0.82, respectively.^{17 18} The scales used are listed in table 1.

This study adopted the critical value calculation method proposed by Li, which is widely recognised in China.¹⁹ The data for the three dimensions of burnout were divided into three equal parts according to the order of high and low, and the boundary value of above one-thirds was taken as the critical value. Burnout was

Table 1 MBI-GS (Chinese version)

Dimensions	Items
Emotional exhaustion	a. I feel emotionally drained from my work
	b. I feel used up at the end of the day
	c. I feel tired when I get up in the morning and have to face another day at work
	d. Working with people all day is a real strain for me
	e. I feel burned out from my work
Depersonalisation	a. I have become more callous towards work since I took this job
	b. I have become less enthusiastic about my work
	c. I doubt the significance of my work
	d. I have become more and more indifferent in the contribution of my job
Personal accomplishment	a. I deal effectively with the problems of clients
	b. I feel that I am contributing to my company
	c. In my opinion, I am good at my job
	d. I feel very happy when I accomplish some tasks of my job
	e. I have accomplished many worthwhile things in this job
	f. I am confident that I can accomplish all tasks effectively
MBI-GS, Maslach Burnout Inventory General Survey.	

categorised into four levels: not detected (scores on all three factors were below the threshold), mild burnout (scores on one factor were equal to or higher than the threshold), moderate burnout (scores on two factors were equal to or higher than the threshold) and high burnout (scores on all three factors were equal to or higher than the threshold).¹⁹

Employee job satisfaction questionnaire

The employee job satisfaction questionnaire was used as a hospital satisfaction questionnaire. The questionnaire was designed according to the requirements of the fourth round of grade hospital accreditation in Zhejiang Province, China, and included eight dimensions. They were compensation and benefits (5 items), development and promotion (4 items), job content and environment (10 items), duty conditions (2 items), supervisory relationships (3 items), peer relationships (4 items), respect and caring (3 items) and sense of belonging (2 items), for a total of 33 items. A five-point Likert scale (ie, from very dissatisfied (1) to very satisfied (5)) was used for each item. Hospitals collect such data annually, which can be

Table 2 Employee job satisfaction questionnaire

Dimensions	Items
Compensation and benefits	Your work efforts are fairly rewarded.
	This hospital will raise your salary from time to time with short intervals.
	Compared with the same type of hospital, the salary of this hospital is relatively high.
	You are satisfied with the benefits (including vacation, training, etc.) other than your salary.
	Compared with other hospitals of the same type, the hospital's benefits are relatively good.
Development and promotion	There are many opportunities for advancement in your job position.
	Those who perform well in their jobs are able to get a fair chance of promotion.
	There are more opportunities for advancement at this hospital compared with similar hospitals.
	You are satisfied with your promotion opportunities (career development).
Work content and environment	Your work gives you a sense of pride.
	Your workplace is comfortable and easy to work efficiently.
	Your work often makes you tired and exhausted.
	You can learn a lot about business and management on the job.
	Your work has been very stressful lately.
	Hospitals are regularly educated on laws and regulations.
	Logistics staff are 'patient centred' and 'staff centred' to meet the needs of medical services.
	The hospital has a well-established logistical supply system to meet the hospital's needs.
	The hospital regularly trains its staff on dispute prevention and handling.
	The hospital is equipped with safety protection facilities to protect the safety of employees.
	The hospital's duty conditions are satisfactory.
Duty condition	The hospital provides good meals for its staff, especially those on night shifts.
Supervisory relationships	Your immediate supervisor trusts you.
	When you experience a grievance at work, the hospital's resolution process is appropriate.
	Some of the hospital's rules and procedures actually make work more difficult.
Peer relationship	You like your coworkers.
	You often put in extra effort because your coworkers are not up to the task.
	You seldom has altercations and conflicts with your coworkers at work.
	You communicate well with your colleagues at the same level.
Respect and caring	The hospital has a department dedicated to receiving complaints from employees.
	Hospitals disclose information to the public and employees to safeguard their democratic rights.
	Hospital leaders report regularly to staff and staff participate in the management of the hospital.
Sense of belonging	On the whole, you like working in this hospital.
	You have never considered leaving this hospital.

used to predict burnout over time, and the questionnaire can be found in [table 2](#).²⁰

Prediction methodology

The most classical four types of machine-learning models were selected, including logistic regression (LR), K-nearest neighbour (KNN), decision tree (DT) and random forest (RF). All models were run under Python (V.3.8) and edited with PyCharm (V.2020.2.2) using the appropriate package.

Logistic regression

The LR model belongs to supervised learning in machine learning, where the dependent variable is a categorical

variable and the independent variables can be categorical or continuous. The principle is to assume that the data information obeys the distribution and then use the maximum likelihood method for parameter estimation. The advantages of the LR model are that it is simple to implement, highly interpretable and widely used in medicine and clinics. The disadvantages are that it has low prediction accuracy and is not applicable to non-linear problems.²¹

K-nearest neighbour

The KNN model, which is widely used in classification problems and belongs to supervised learning, is one of

the most mature and simple machine-learning models. The core principle is to determine a new target based on features by finding the K cases closest to the predicted target and grouping the features of these cases into a class. KNN has the advantage of being simple, efficient and widely applied, but has the disadvantage of low interpretability and poor ability to analyse unbalanced samples.²²

Decision tree

A DT model is a machine-learning model that typically solves classification problems. DT uses a set of rules to classify data so that predictions can be made regarding unknown data based on these rules. The advantages of DT are that it is easy to understand, interpret, visualise and improve the rules. However, the disadvantage of the DT is that the model is prone to overfitting.²³

Random forest

RF is based on DT and contains multiple DTs such that the prediction target can be viewed from multiple perspectives to obtain a group opinion. Therefore, RF has good accuracy and avoids overfitting due to the introduction of randomness, is suitable for a wide range of problems and is widely used in healthcare, clinical, insurance and market analyses. However, if it contains too many DTs, it may lead to a slower model.²⁴

K-fold cross-validation method

To reduce the risk of overfitting or underfitting, as well as generalisation errors, the K-fold cross-validation method was used. The K-fold cross-validation method divides all the data in the original dataset into K subsamples, selects one of the samples as the test set without repetition and the remaining K-1 samples as the training set. It repeats K times and takes the average value of the evaluation indices obtained from the K times to obtain the evaluation results of the model. Since the tenfold cross-validation method is the most common, the tenfold cross-validation method is also used in this study.

Regarding the evaluation metrics of the prediction methods used in this study, the dependent variable, burnout, is a four-categorical variable suitable for categorisation algorithms. Therefore, the accuracy and F1-score based on the confusion matrix were selected as the model evaluation metrics. Accuracy refers to the proportion of the correctly predicted sample size to the total predicted sample size; a value closer to 1 indicates a higher accuracy of the model. The F1-score is the weighted average of model precision and recall, where precision refers to the probability of a correct prediction among samples predicted to be positive, and recall is the proportion of samples that are actually positive and considered to be positive. To reconcile precision and recall, scholars have proposed the F1-score, which is calculated as follows:

$$F1 = \frac{2PR}{P+R}$$

where P is the precision rate and R is the recall rate.²⁵

Because there were four categories in this study, their F1-scores were calculated as precision and recall for each category separately to obtain their respective F1-scores and then averaged to obtain the total F1-score, that is, macro-F1.

RESULTS

Reliability and validity test

In this study, the minimum value of Cronbach's alpha for the three dimensions of the MBI-GS was 0.945. The minimum Cronbach's alpha value for the eight dimensions of the job satisfaction questionnaire was 0.738. The alpha for all dimensions of both questionnaires was greater than 0.7, and the reliability of the questionnaire met the requirements.

In accessing validity, this study employed factor analysis. The Kaiser-Meyer-Olkin (KMO) test yielded a KMO value of 0.92 for the MBI-GS. Concurrently, Bartlett's test of sphericity returned a significant p value of <0.001. The KMO test results of the employee job satisfaction questionnaire showed a KMO value of 0.977, while Bartlett's spherical test results showed a significant p value of <0.001. The validity of both questionnaires was significant; there was a correlation between the variables, the factor analysis was valid and there was good reliability.

Burnout status

Excluding the blank and abnormal data, the remaining effective sample size was 1125 cases, which met the requirement of the sampling sample size (not less than 369). The proportion of females in this study was 83.3%, and 52.4% were between 30 and 39 years of age. In terms of occupational category, 58.3% were nurses and 20.9% were doctors. Academic qualifications were mainly bachelor's degrees, with a balanced distribution of working years, and titles were mainly junior and intermediate. Further details are presented in online supplemental file 1.

Burnout thresholds were calculated using the method proposed by Li *et al*,¹⁹ with a critical value of 16 for the EE dimension, 8 for the DP dimension and 23 for the PA dimension. Burnout was observed in 61.2% of hospital staff, with 19.8% experiencing moderate burnout and 9.8% experiencing high burnout. Employee satisfaction was assessed by calculating the mean value of each dimension, with the average satisfaction score for each dimension being close to 4.

Relationship between burnout levels and variable

There were significant differences ($p < 0.05$) in age, occupational group, area classification, work experience and job series for different burnout levels. The data showed higher rates of high burnout in the 30-39 age group, physicians, surgery and 0-5 work years. The results are summarised in online supplemental file 2.

Furthermore, there was a significant correlation ($p < 0.05$) between hospital staff and the variables in each dimension of satisfaction. The least significant difference

Table 3 Burnout-level results (satisfaction)

Satisfaction variable	Mean	SD	F	P	LSD
Compensation and benefits					
A Not detected	4.23	0.592	185.047	<0.001	A>B>C>D
B Mild burnout	3.839	0.802			
C Moderate burnout	3.483	0.752			
D High burnout	2.564	0.63			
Development and promotion					
A Not detected	4.388	0.553	206.754	<0.001	A>B>C>D
B Mild burnout	4.084	0.757			
C Moderate burnout	3.656	0.766			
D High burnout	2.714	0.549			
Job content and environment					
A Not detected	4.35	0.458	238.708	<0.001	A>B>C>D
B Mild burnout	4.032	0.591			
C Moderate burnout	3.746	0.57			
D High burnout	2.934	0.376			
Duty conditions					
A Not detected	4.449	0.552	219.299	<0.001	A>B>C>D
B Mild burnout	4.145	0.801			
C Moderate burnout	3.717	0.675			
D High burnout	2.705	0.633			
Superior–subordinate relationship					
A Not detected	4.391	0.428	387.401	<0.001	A>B>C>D
B Mild burnout	3.792	0.61			
C Moderate burnout	3.44	0.452			
D High burnout	2.797	0.439			
Peer relationship					
A Not detected	4.546	0.4	369.943	<0.001	A>B>C>D
B Mild burnout	4.273	0.508			
C Moderate burnout	4.017	0.46			
D High burnout	2.991	0.392			
Respect and care					
A Not detected	4.652	0.448	479.753	<0.001	A>B>C>D
B Mild burnout	4.317	0.609			
C Moderate burnout	3.753	0.516			
D High burnout	2.725	0.385			
Sense of belonging					
A Not detected	4.613	0.457	559.797	<0.001	A>B>C>D
B Mild burnout	4.261	0.604			
C Moderate burnout	3.455	0.43			
D High burnout	2.773	0.343			
LSD, least significant difference.					

LSD, least significant difference.

test revealed significant differences in satisfaction scores for different levels of burnout. Burnout was strongly correlated with the dimensions of case-hospital satisfaction (table 3).

Multiple LR for burnout

Since the dependent variable, burnout level, was a four-categorical variable, multiple LRs were chosen for multifactorial analysis. Because of the strong intrinsic

Table 4 DT and RF model characteristics weights

Characteristics	Weights of DT characteristics	Weights of RF characteristics
Sense of belonging	29.20%	20.70%
Supervisor–subordinate relationship	20.40%	18.20%
Respect and care	6.50%	12.30%
Peer relationship	13.70%	10.60%
Development and promotion	5.50%	8.50%
Work content and environment	9.10%	6.90%
Compensation and benefits	6.10%	5.90%
Duty condition	1%	5.10%
Working experience(ref=>15)	1.10%	1.80%
BMI	3%	1.50%
Job series	0	1.50%
Area classification	3.50%	1.40%
Age(ref=>50)	0.90%	1.20%
Occupational group	0	1.10%
Patient facing	0	1.00%
Education	0	1.00%
Title	0	0.80%
Gender	0	0.50%

BMI, body mass index; DT, decision tree; RF, random forest.

correlation among occupational group, area classification and job series, only the occupational group was chosen to enter the model. The model-dependent variable burnout level with 'not detected' as the reference category showed a pseudo-R-squared Cox–Snell value of 0.723. The area under curve value is 0.917. The model was significant at the overall level ($p<0.05$), making it valid.

Online supplemental file 3 shows that, in mild burnout, compensation and benefits, development and promotion, job content and environment, duty conditions and supervisor–subordinate relationships were significantly correlated ($p<0.05$) with the level of burnout. In addition to age, which also showed a significant correlation ($p<0.05$) with the level of burnout.

In moderate burnout, job content and environment, sense of belonging, respect and care, peer relationships, duty conditions and superior–subordinate relationships showed significant correlations ($p<0.05$) with burnout levels. And nurses in the occupational group showed significant correlations ($p<0.05$) with burnout levels.

In high burnout, there is a significant correlation ($p<0.05$) between job content and environment, sense of belonging, respect and care, peer relationships and superior–subordinate relationships. And nurses in

occupational groups are also significantly correlated ($p<0.05$) with burnout level.

Machine-learning model results

In this study, the LR, KNN, DT and RF models were used; 70% of the dataset was used as the training set, and 30% was used as the test set. To avoid overfitting, all models were validated using the tenfold crossover method.

The results showed that the LR model test set prediction accuracy was 0.713, the F1-score was 0.703 and the model predicted well. The LR model cross validation achieved an accuracy of 0.685 and an F1-score of 0.682. The prediction accuracy of the KNN model test set was 0.592, and the F1-score was 0.57. An accuracy of 0.624 was obtained using tenfold cross validation with an F1-score of 0.603, which was slightly lower than that of the LR prediction. The DT model test set prediction accuracy was 0.757, and the F1-score was 0.751 and calculated the characteristic weights for each variable. Tenfold cross validation achieved an accuracy of 0.732 with an F1-score of 0.727, which is a significant improvement over the LR and KNN model predictions. The RF model test set prediction accuracy was 0.805, and the F1-score was 0.802 and calculated the characteristic weights for each variable (table 4). Using tenfold cross validation, we achieved an accuracy of 0.79 and an F1-score of 0.781, which is the highest prediction accuracy among the four classes of algorithmic models (table 5).

DISCUSSION

Burnout characteristics

The survey of burnout levels in hospital revealed that 61.2% of the medical staff exhibited one or more burnout symptoms. This study found that burnout was lower in hospitals for people older than 50 years, who were likely to have higher levels of seniority and title, a higher sense of achievement and relatively strong relationships, and scored lower on emotional exhaustion. In terms of occupational group, area classification and job series, all three indicated that burnout was somewhat higher in the healthcare group than in the non-healthcare population, which is consistent with See *et al*'s study.²⁶ Furthermore, burnout is somewhat higher among clinical healthcare workers than among medical technicians and administrative staff. Related studies have shown that frontline clinical healthcare workers experience higher burnout.²⁷ Additionally, this study found a significant correlation between hospital burnout and the dimensional variables of employee satisfaction, which is consistent with Mohr *et al*'s study.²⁸ This implies that increasing employee satisfaction among medical staff is important for reducing medical burnout.

High burnout can significantly increase the propensity to leave,²⁹ jeopardising the physical and psychological well-being of employees and the quality and safety of healthcare.^{30 31} Nearly, 9.8% of the hospital personnel had high burnout. The rate of high burnout was highest

Table 5 Machine-learning model evaluation results

Characteristics	Accuracy	Recall rate	Precision rate	F1
LR				
Training sets	0.719	0.719	0.717	0.718
Test set	0.713	0.713	0.703	0.703
Cross-validation set	0.685	0.685	0.693	0.682
KNN				
Training sets	0.696	0.696	0.698	0.681
Test set	0.592	0.592	0.578	0.57
Cross-validation set	0.624	0.624	0.624	0.603
DT				
Training sets	0.873	0.873	0.879	0.87
Test set	0.757	0.757	0.762	0.751
Cross-validation set	0.732	0.732	0.742	0.727
RF				
Training sets	0.889	0.889	0.895	0.887
Test set	0.805	0.805	0.812	0.802
Cross-validation set	0.79	0.79	0.794	0.781

DT, decision tree; KNN, K-nearest neighbour; LR, logistic regression; RF, random forest.

in the 30–39 age group, at approximately 11%. This may be due to higher levels of burnout resilience in older physicians.³² In the occupational group, doctors and nurses burnout rates are higher relative to other hospital employees, which is consistent with previous research.³³ Regarding ward classification, physicians and surgeons had the highest rates of burnout, with 11%. Regarding years of experience, 0–5 years and 5–10 years had the highest rates of high burnout. These characteristics reflect the high burnout population in Chinese hospitals. That is, medical students, after years of medical education, face technical, financial, self-doubt and responsibility pressures in the first 10 years of their careers, which puts them under tremendous pressure and produces high burnout.^{34 35} This group of people requires guidance, attention and help from hospitals and related organisations.

Evaluation of machine-learning model prediction effectiveness

The results of the tenfold cross validation of the four types of models, LR, KNN, DT and RF, showed that RF had the highest accuracy and F-value, followed by DT and LR, while KNN had the lowest accuracy and F-value. The present study concluded that RF is a more appropriate model for predicting burnout among medical personnel.

The prediction results of this study show that all four types of machine-learning models can predict hospital staff with high burnout to some extent. The RF model can achieve 80% accuracy and avoid overfitting and underfitting risks to a certain extent, which has a strong application value. This may be because the RF fits human psychological and behavioural predictions. For instance,

Chae *et al* used the RF to predict participation in a telephone social contact intervention for socially isolated older adults, which worked well.³⁶ Kim *et al* used RF to predict employee depression and achieved the highest prediction accuracy.³⁷ Burnout, on the other hand, is closely related to occupational psychology and behaviour, so it fits better. Therefore, we believe that RF may have a wider scope of application in the fields of psychological behaviour and occupational health.

In the final RF prediction model, we found that the dimensions of job satisfaction accounted for a very high percentage of burnout prediction weights. This is similar to previous studies, such as Song *et al*, who found that job satisfaction was a negative predictor of burnout (direct effect of –0.684).³⁸ Quesada-Puga *et al* concluded that the lower the job satisfaction of nurses, the higher the level of burnout.³⁹ Tu *et al*'s study concluded that job rewards satisfaction, personal development satisfaction, work internal environment satisfaction and burnout have a significant correlation.⁴⁰ van der Doef *et al* noted that improved working conditions could reduce high levels of burnout and physical complaints among East African nurses.⁴¹ And in the field of occupational therapists, Park's study concluded that job satisfaction is significantly negatively correlated with burnout in terms of effect size.⁴² However, previous studies have tended to look at predictions and correlations between the two, few studies to analyse the causal relationship between job satisfaction and burnout, which needs more research.

Furthermore, since employee satisfaction data in the hospital were collected annually, it is possible to predict the corresponding burnout data on an annual basis, thus

realising a sustainable prediction of hospital employee burnout. Additionally, hospitals can collect a portion of their burnout data each year for training and learning to achieve better predictions. This forecasting mechanism can assist managers in gaining a comprehensive understanding of medical staff burnout to adjust their work-force strategies. It can also focus on employees who may need assistance with improvement methods. This reduces the negative impact of burnout on turnover and the quality of medical care. The unique contribution of this study compared with the existing research is the application of several machine-learning methods for more accurate prediction of medical staff burnout. This study can assist hospital administrators in identifying staff at high risk of burnout early on and taking steps to improve the situation.

Strengths and limitations of the study

This study analysed the current state of burnout in hospitals and identified a machine-learning model that is more suitable for burnout prediction by investigating burnout in hospitals and synthesising employee demographic characteristics and satisfaction. This study identified the main characteristics of highly burned medical staff groups, established a preliminary burnout early warning system for hospital medical staff and provided a reference basis and new ideas for burnout management.

One of the limitations of this study is that there is still a lack of large-sample and authoritative studies on the critical value and level division method of burnout in China. This, on the one hand, leads to the fact that the prediction model of this study is only applicable to the hospital in this study, while the criticality and division need to be recalculated for different hospitals in different regions, and also that the four-classification division method interferes with the accuracy of the prediction method to a certain extent. This study requires more data and a longer time period to validate the accuracy of the prediction model. In addition, more research is needed to determine where hospitals should start to improve after identifying highly burned-out personnel.

CONCLUSION

The study revealed that 61.2% of the medical staff exhibited at least one symptom of burnout. A significant correlation was found between job satisfaction and burnout levels. Notably, physicians and surgeons with less than a decade in the profession were more prone to high burnout. For predictive purposes, the RF model proved effective in predicting the burnout level among medical staff, consistently achieving an accuracy rate close to 80%. These findings can assist hospital administrators in both preventing and mitigating burnout among medical staff.

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Patient consent for publication Not applicable.

Ethics approval This study involves human participants. All procedures adhered to the guidelines of the Institutional Review Board of Taizhou Hospital of Zhejiang Province, affiliated with Wenzhou Medical University (approval number: K20230615) and aligned with the tenets of the Declaration of Helsinki. All participant's information were collected anonymously. Participants gave informed consent to participate in the study before taking part.

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